

2 MARKS QUESTION ANSWERS

Q1. Define AI. Explain how it differs from conventional programming.

AI is the field of making machines perform tasks that need human intelligence, like learning and decision-making.

- AI systems learn from data and adapt; conventional programs follow fixed instructions only.
- AI can handle uncertain or new situations; conventional programming works only for pre-defined cases.

Q2. Why is search considered an important technique in AI? Give a suitable example.

Search means exploring possible solutions step by step to find the best one.

- Many AI problems have no direct formula, so search helps find correct solutions systematically.
- Example: Finding the shortest path in Google Maps uses search techniques.

Q3. Explain the importance of abstraction as an AI problem-solving technique.

Abstraction means ignoring unnecessary details and focusing only on important features of a problem.

- It simplifies complex real-world problems into manageable models for the system.
- It reduces computation time by removing irrelevant data from consideration.

Q4. What are the main approaches of AI? Give a real-life example.

AI has different approaches based on how intelligence is achieved.

- Main approaches: Symbolic (rule-based) AI and Connectionist (neural network based) AI.
- Example: Symbolic AI is used in expert systems; connectionist AI is used in face recognition apps.

Q5. Define cognitive science.

Cognitive science is the study of how the human mind thinks, learns, and solves problems.

- It combines psychology, neuroscience, and computer science to understand intelligence.
- It helps AI researchers design systems that mimic human thinking.

Q6. What is the main aim of cybernetics?

Cybernetics is the study of control and communication in machines and living beings.

- Its aim is to understand how systems use feedback to control their behavior.
- It helped build the foundation for automation and robotics in AI.

Q7. What are the major components and goals of AI?

AI systems are built using key components to achieve intelligent behavior.

- Major components: Knowledge representation, reasoning, learning, and perception.
- Goals of AI: To build systems that can think, learn, and act like humans efficiently.

Q8. Differentiate between formal tasks and mundane tasks in AI.

Formal Tasks	Mundane Tasks
Based on fixed rules and logic	Based on everyday human activities
Example: Mathematics, games like chess	Example: Seeing, walking, talking
Easier for computers to perform	Harder for computers to perform
Follow clear, defined steps	Involve uncertainty and common sense

Q9. Define state-space representation.

State-space representation is a method to represent all possible states of a problem and the moves between them.

- It includes initial state, goal state, and operators to move between states.

- It helps in systematically searching for a solution path.

Q10. Define the water jug problem.

The **water jug problem** is a classic AI problem where jugs of different capacities are used to measure an exact amount of water.

- It has no measuring markers, so operations like fill, empty, and pour are used.
- It is solved using state-space search to reach the goal amount of water.

Q11. Differentiate between monotonic and non-monotonic production systems.

Monotonic System	Non-Monotonic System
Application of one rule does not affect other rules	Application of one rule can affect other rules
Knowledge base only grows, never shrinks	Knowledge base can change or reduce
Simpler and more predictable	More complex and flexible
Used when facts are fixed and certain	Used when facts can be revised later

Q12. Explain brute-force strategies in AI.

Brute-force search explores all possible solutions without using extra knowledge about the problem.

- Examples include Breadth-First Search (BFS) and Depth-First Search (DFS).
- It guarantees a solution but is slow and uses more memory for large problems.

Q13. What is a heuristic function? Give an example.

A **heuristic function** estimates the cost or distance from the current state to the goal state.

- It helps search algorithms choose the most promising path quickly.
- Example: Straight-line distance to destination used in map navigation.

Q14. Define a plateau in AI search.

A **plateau** is a flat area in the search space where neighboring states have equal heuristic values.

- It makes it difficult for the algorithm to decide the next best direction.
- It can cause hill climbing search to stop or wander without reaching the goal.

Q15. Differentiate between Breadth-First Search (BFS) and Best-First Search.

BFS	Best-First Search
Explores all nodes level by level	Explores the most promising node first
Does not use heuristic values	Uses heuristic function to guide search
Guarantees shortest path in unweighted graphs	Does not always guarantee shortest path
Slower for large search spaces	Faster due to informed guidance

Q16. Differentiate between A* and AO* search algorithms.

A* Algorithm	AO* Algorithm
Used for OR graphs	Used for AND-OR graphs
Finds a single best path to goal	Finds a solution tree, not just a path
Uses $f(n) = g(n) + h(n)$	Uses cost of AND-OR combined subgoals
Simpler and used in pathfinding	Used in complex problems with sub-problems

Q17. What are the major components of a game-playing program?

A game-playing program needs several key parts to make intelligent decisions.

- Components: Initial state, set of legal moves, terminal test, and utility function.

- It also uses a search algorithm like minimax to choose the best move.

Q18. What is the advantage of alpha-beta pruning over the minimax algorithm?

Alpha-beta pruning is an optimization technique applied to the minimax algorithm.

- It skips branches that will not affect the final decision, saving time.
- It reduces the number of nodes evaluated, making search faster than plain minimax.

Q19. What is a game of chance? Give a real-life example.

A **game of chance** is a game where the outcome depends partly on randomness, not just player skill.

- It uses probability along with search techniques like expectimax to decide moves.
- Example: Games like backgammon or card games that involve dice or shuffling.

Q20. Consider an intelligent medical diagnosis system that receives symptoms from a patient and recommends possible diseases. Explain how Knowledge Representation, Search, Reasoning, NLP, and Expert Systems can contribute to the system. Which component is most popular and why?

- Knowledge Representation stores disease-symptom relationships; Search finds matching disease possibilities from symptoms.
- Reasoning draws conclusions from facts and rules; NLP understands patient symptoms given in natural language; Expert Systems combine all this to give diagnosis and advice like a human doctor.

Expert Systems is the most popular component, because it directly integrates knowledge, reasoning, and rules to give practical, human-like medical decisions.